

Answer each question in your workbook. Show your working.

## Method

Ratio  $r = \text{term} \div \text{previous term}$     Check  $r$  is the same throughout    Next term = current  $\times r$      $n$ th term = first  $\times r^{(n-1)}$

Check against the pattern

## Common mistakes

- Adding instead of multiplying.
- Dividing the wrong way to find  $r$ .
- Decreasing sequences have a fractional ratio.
- Using  $r^n$  instead of  $r^{(n-1)}$ .

## Worked example

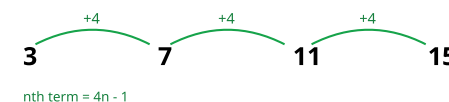
Continue 2, 6, 18, 54, ...

$$r = 6 \div 2 = 3. \text{ Next} = 54 \times 3 = 162.$$

$$5\text{th term} = 2 \times 3^4 = 162.$$

## Visual aid

Visual Example



In a geometric sequence each term is the previous term multiplied by the same ratio  $r$ . With  $r > 1$  the terms grow quickly; with  $0 < r < 1$  they shrink towards zero.

# Questions

Answer each question in your workbook. Show your working.

**Q1 of 8**

[2] ● ○ ○ ○ ○

Write the next two terms: a) 3, 6, 12, 24, ... b) 1, 5, 25, 125, ...

**Q2 of 8**

[2] ● ○ ○ ○ ○

Find the common ratio: a) 4, 8, 16, ... b) 2, 6, 18, ...

**Q3 of 8**

[3] ● ● ○ ○ ○

Write the next two terms: a) 5, 15, 45, ... b) 80, 40, 20, ... c) 1, -2, 4, -8, ...

**Q4 of 8**

[3] ● ● ○ ○ ○

Find the term: a) first 3,  $r = 2$ , 4th term b) first 2,  $r = 5$ , 3rd term c) first 100,  $r = \frac{1}{2}$ , 4th term

**Q5 of 8**

[3] ● ● ● ○ ○

Arithmetic or geometric (give  $d$  or  $r$ )? a) 2, 4, 6, 8 b) 2, 4, 8, 16 c) 5, 10, 20, 40

**Q6 of 8**

[3] ● ● ● ○ ○

Bacteria double each hour, start at 50. a) Number after 0, 1, 2, 3 hours. b) After 5 hours?

**Q7 of 8**

[4] ● ● ● ● ○

Geometric, first term 4, ratio 3. a) First four terms. b) 6th term. c) Which term is 2916?

**Q8 of 8**

[5] ● ● ● ● ●

Ball dropped from 20 m, bounces to 0.8 of previous height. a) Heights after 3 bounces. b) Arithmetic or geometric? c) Will it ever reach 0? Explain.